



Department of Computer Engineering

Faculty of Engineering

University of Sri Jayewardenepura

Course	Software Engineering
Course Code	CO2307
Title	UML Diagrams (Reschedule)
Time	1 Hour
Outcomes	Getting Basic Knowledge UML Diagrams

1. Use Case Diagram- Hospital Appointment Booking System

A private hospital wants to digitize its appointment process. Patients can register online, search for available doctors by specialty, and book or cancel appointments. A doctor can view their schedule and update their availability. A receptionist can book appointments on behalf of walk-in patients. The system sends automatic reminders to patients before their appointment. Admin can generate reports on appointment statistics.

2. Activity Diagram - Student Course Registration System

At the start of each semester, students at a university must register for courses online. The student logs into the portal, browses available courses, and checks prerequisites. If prerequisites are met, the student adds the course to their cart. Once selection is complete, the system checks for seat availability. If seats are available, the registration is confirmed and a fee invoice is generated. If not, the student is added to a waitlist. Payment must be completed within 48 hours, or the registration is automatically cancelled.

3. Sequence Diagram - Food Delivery Mobile Application

A customer opens a food delivery app and logs in. The app authenticates the user with the backend server, which verifies credentials from the database. The customer searches for a restaurant, selects items, and places an order. The system confirms item availability with the restaurant, processes payment through a payment gateway, and sends an order confirmation to both the customer and the restaurant. The restaurant then updates the order status to "Preparing."

